

CLIFFE OWINO IBANDE

0790382913 | owinocliffe10@gmail.com | github.com/thestartupdeveloper10 | <https://cliffe-ibande.onrender.com/>

EDUCATION

Kisii University, Kisii County, Kenya

Bachelor of Science in Software Engineering

2020 – Dec 2024

Maseno School, Kisumu County, Kenya

Kenya Certificate of Secondary Education

2016 – Dec 2020

CERTIFICATIONS & ADDITIONAL COURSES

- Cloud Architecture & DevOps Essentials – AWS/Azure (2023)
- SPSS for Data Analysis (2024)
- Huawei Cloud Fundamentals (2023)

TECHNICAL SKILLS

Languages:

Python, JavaScript, Java, TypeScript, HTML, CSS, SQL, Bash

Web & Mobile Frameworks:

Node.js, React, Frappe, Next.js, Django, Flask, React Native, SpringBoot, Material-UI, GraphQL

Developer Tools:

Git, Docker, Kubernetes, Jenkins, VS Code, PyCharm, IntelliJ, Eclipse, Linux (Ubuntu/CentOS), Cron jobs, Shell Scripting, NGINX, Virtualization (VMware, VirtualBox)

Libraries & AI:

Pandas, NumPy, Matplotlib, Seaborn, scikit-learn, TensorFlow, PyTorch, LangChain, Hugging Face Transformers

Cloud & DevOps:

Huawei Cloud, AWS, Azure, Google Cloud Platform
Microservices, CI/CD pipelines, REST & gRPC services
Grafana, Prometheus, Loki (Monitoring & Logging)

Databases:

MongoDB, PostgreSQL, MySQL, Redis

Data Analysis Tools

SPSS, Excel Analysis, Python Scientific Stack

EXPERIENCE

Operations Engineer Intern

Prodapt Ltd
September 2025 – Present
Supporting M-Pesa operations across 5 nations: Mozambique (MZ), Lesotho (LS), Tanzania (TZ), Ghana (GH), and Democratic Republic of Congo (DRC).
Working under the Core Payment Services (CPS) division, ensuring reliability, availability, and security of mobile money platforms.

Key Responsibilities:

- Managed and maintained large-scale Linux-based production servers supporting M-Pesa's core payment infrastructure.
- Worked extensively with the Huawei technology stack, including cloud services, monitoring tools, middleware, and network components.
- Performed database monitoring, tuning, troubleshooting, and optimized queries for high-volume transactional systems.
- Handled VM provisioning, virtualization management, and performance tracking across distributed infrastructures.
- Executed certificate creation, renewal, and deployment for secure communication between payment services.

- Diagnosed and resolved network-related issues, including latency, routing, connectivity, and service availability problems.
- Conducted data analysis and reporting to identify system bottlenecks, anomalies, and operational irregularities.
- Collaborated with cross-functional teams to support incident management, service restoration, and root-cause analysis.
- Utilized monitoring tools (Grafana, Digital View) to ensure system health and SLA compliance.
- Automated repetitive tasks using Python, Bash, and internal scripts, improving efficiency and reducing manual errors.

Backend Web Developer

Techsavannah

Sep 2024 – July 2025

- Delivered bespoke software solutions and IT consulting services for diverse industry clients.
- Integrated microservices architectures, Dockerized applications, and Kubernetes deployments for scalable cloud environments.
- Implemented Linux-based server automation, cron jobs, monitoring dashboards, and deployment pipelines.
- Collaborated with cross-functional teams to optimize business processes and modernize legacy systems.
- Developed high-availability systems with API gateways, reverse proxies, and load balancing (NGINX).

Undergraduate Industrial Attachment (Data analyst)

Huawei Tech. (Kenya) CO., LTD

June 2023 – Aug 2023

- Designed and refined data models for efficient data organization and management.
- Built interactive dashboards improving visualization and strategic decision-making.
- Assisted in constructing and maintaining data cubes for analytics.
- Generated advanced network monitoring reports and performance insights.
- Conducted data analysis using SPSS, Python, and Excel.

- Exposure to cloud architecture, virtualization, cloud monitoring, and distributed systems.

PROJECTS

Fullstack & DevOps Projects

Linux-Based Monitor & Microservice Deployment

October 2025 – Present

Linux, Docker, Kubernetes, Prometheus, Grafana, Microservices (FastAPI/Node.js)

- Built a production-ready microservice application deployed on Docker containers.
- Orchestrated services using Kubernetes with horizontal scaling and health checks.
- Implemented monitoring using Prometheus + Grafana with custom metrics.
- Automated deployments using shell scripts and cron jobs.

RAG-Based Document Intelligence System

September 2025 – October

Python, LangChain, Hugging Face, FastAPI, MongoDB, Docker

- Built a Retrieval-Augmented Generation (RAG) pipeline for intelligent search and document processing.
- Integrated vector databases for high-speed semantic search.
- Deployed scalable inference pipelines using Docker.
- Used FastAPI to expose LLM endpoints with secure token auth.

NGINX SSL Automation & Certificate Management

Jan 2025 – Feb

Linux, NGINX, OpenSSL, Certbot, Bash

- Implemented automated SSL certificate creation and renewal using Certbot and cron.
- Configured NGINX reverse proxy for secure microservices traffic routing.

Fullstack Web Development

Motimarket – Car Sales Platform

React (TypeScript), SpringBoot REST Framework, PostgreSQL

October 2025 – Present

- Developed a full-featured car sales web application with comprehensive listings and advanced search filters.
- Implemented secure user authentication and efficient data management using Spring REST Framework and PostgreSQL.
- Focused on a modern UI/UX design to enhance user engagement and conversion rates.

The Board FC

React (Typescript), Supabase, Tanstack query, Email integration

August 2025 – September

- A fully functional modern sports web app featuring all the use cases for a football website.
- It includes merchandise store, admin dashboard, a blog and a fan zone area.

Label safi

MERN Stack (MongoDB, Express.js, React, Node.js), Redux, Tailwind CSS

Feb 2025 – August

- A premium e-commerce platform showcasing exclusive designer shoes and clothing collections.
- Features a sophisticated shopping experience with real-time inventory management, personalized recommendations, and seamless checkout process.
- The platform combines luxury fashion with modern e-commerce functionality.

Royal Watches

MERN Stack (MongoDB, Express.js, React, Node.js), Tailwind CSS

August 2024 – September 2024

- Developed a dynamic e-commerce website featuring product listings, secure user authentication, and integrated payment gateways.
- Leveraged GraphQL for efficient data querying and Tailwind CSS for a modern, responsive design.

SkillUp

Python, Flask, React, SQL, Docker, React Native, K-means, Git

May 2024 – August

- Created a machine learning-based web application that analyzes skill gaps to align workforce development.
- Integrated modern RESTful APIs with React Native for seamless data visualization and collaboration tracking.
- Expanded functionalities to include personalized training recommendations using AI.

Shutter Spot

MERN Stack (MongoDB, Express.js, React, Node.js), Google Maps API, Postgres, socket io

Feb 2024 – Apr 2024

- Built a platform connecting photographers with a global clientele using location-based services.
- Integrated Google Maps API to enhance location discovery and streamline client-photographer interactions.
- Delivered a user-friendly interface focused on simplicity and responsiveness.

AI, Machine Learning & LLM Projects

SkillUp AI Enhancement

Python, Flask, React, SQL, Docker, K-means, Hugging Face Transformers

June 2025 – August

- Integrated large language models into the SkillUp platform to enhance natural language processing and data analysis.
- Developed clustering algorithms and personalized recommendation engines to drive targeted training initiatives.

ConversaAssist – Conversational AI Chatbot

Python, Flask, Hugging Face Transformers, Docker

April 2025 – May

- Developed an intelligent conversational agent for customer support that leverages state-of-the-art LLM APIs.
- Designed and implemented RESTful endpoints for real-time query handling and FAQ resolution.
- Integrated containerization for scalable deployment, reducing response times and enhancing user engagement.

SmartDoc Summarizer – Automated Document Summarization Tool

Python, FastAPI, Hugging Face Transformers, PostgreSQL

Jan 2025 – March

- Built a web service that processes lengthy documents and generates concise summaries using pre-trained LLMs.
- Implemented API endpoints to handle document uploads and summary retrieval, streamlining information consumption.
- Optimized summarization accuracy and performance through iterative testing and model fine-tuning.

Cloud Projects

Huawei Cloud Resource Automation

Huawei Cloud ECS, OBS, Cloud Monitoring, Bash, Python

- Automated cloud resource creation and monitoring alerts.
- Integrated container deployments on ECS instances.

Mobile Development

JamiiBudget

React Native, Appwrite, TypeScript

Sept 2024 – Jan 2025

- Developed a comprehensive full-stack expense tracking mobile application with robust financial management features.
- Implemented secure user authentication and real-time analytics with Appwrite as the backend service.

- Enhanced user experience through intuitive interface design and responsive layouts.

Data Science Projects

Cost and Profitability Analysis

Python, Pandas, NumPy, Matplotlib

Jan 2024 – Feb 2024

- Analysed business cost structures and revenue streams to provide actionable financial insights.
- Leveraged Python libraries to manipulate data and visualize key performance indicators.

Dynamic Pricing Strategy

Python, Pandas, NumPy, Matplotlib

Dec 2023 – Jan 2024

- Developed real-time pricing models that adjust based on market trends and competitor pricing.
- Utilized data visualization to identify factors influencing revenue optimization.

ADDITIONAL INFORMATION

- **Networking & AI Enthusiast:** Continuously upgrading skills in advanced networking technologies and AI, with hands-on experience in both traditional and cutting-edge development practices.
- **Professional Growth:** Actively pursuing certifications and courses to stay ahead in cloud computing, DevOps, and emerging technologies like large language models.
- **Collaboration & Innovation:** Passionate about leveraging modern tech stacks to solve real-world problems and drive business transformation.